

PDE Exam Reference Sheet — Ultimate Edition

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Trigonometric & Hyperbolic Identities

Pythagorean: $\sin^2 \theta + \cos^2 \theta = 1$, $1 + \tan^2 \theta = \sec^2 \theta$

Euler / Complex: $e^{i\theta} = \cos \theta + i \sin \theta$, $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$,
 $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

Hyperbolic: $\cosh x = \frac{e^x + e^{-x}}{2}$, $\sinh x = \frac{e^x - e^{-x}}{2}$,
 $\cosh^2 x - \sinh^2 x = 1$

Sum / Difference:

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\sinh(a \pm b) = \sinh a \cosh b \pm \cosh a \sinh b$$

$$\cosh(a \pm b) = \cosh a \cosh b \pm \sinh a \sinh b$$

Double / Half angle: $\sin 2\theta = 2 \sin \theta \cos \theta$, $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$, $\sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$, $\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}$

Product-to-Sum:

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

Taylor / Power Series

$f(x)$	Series	R
e^x	$\sum x^n/n!$	∞
$\sin x$	$\sum (-1)^n x^{2n+1}/(2n+1)!$	∞
$\cos x$	$\sum (-1)^n x^{2n}/(2n)!$	∞
$\sinh x$	$\sum x^{2n+1}/(2n+1)!$	∞
$\cosh x$	$\sum x^{2n}/(2n)!$	∞
$\ln(1+x)$	$\sum (-1)^{n+1} x^n/n$	$ x \leq 1$
$1/(1-x)$	$\sum x^n$	$ x < 1$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \dots$	$ x < 1$
$\arctan x$	$\sum (-1)^n x^{2n+1}/(2n+1)$	$ x \leq 1$
$\tan x$	$x + x^3/3 + 2x^5/15 + 17x^7/315 + \dots$	$ x < \pi/2$
$\coth z$	$1/z + z/3 - z^3/45 + \dots$	$0 < z < \pi$

Useful Limits & Series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, |r| < 1, \quad \int_0^{\infty} \frac{\sin^2 u}{u^2} du = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\sin u}{u} du = \frac{\pi}{2}$$

Orthogonality & Fourier

Standard orthogonality on $[0, L]$

$$\int_0^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = \frac{L}{2} \delta_{mn}, \quad \int_0^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{L}{2} \delta_{mn},$$

$$\int_0^{2\pi} \sin mx \cos nx dx = 0$$

Fourier series on $[-L, L]$

$$f = \frac{a_0}{2} + \sum (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}); \quad a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx$$

Fourier transform (asymmetric convention)

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk$$

Key pairs ($a > 0$): $e^{-ax^2} \leftrightarrow \sqrt{\pi/a} e^{-k^2/(4a)}$; $e^{-a|x|} \leftrightarrow$

$$\frac{2a}{a^2+k^2}; \quad \delta(x) \leftrightarrow 1; \quad \mathcal{F}^{-1}\{(a+k^2)^{-1}\} = (\pi/\sqrt{a}) e^{-\sqrt{a}|x|}$$

Properties: $\widehat{f'} = ik\hat{f}$; $\widehat{f''} = -k^2\hat{f}$; $\widehat{xf} = i\hat{f'}$; $\widehat{f(x-a)} = e^{-ika}\hat{f}$; $\widehat{f * g} = \hat{f} \cdot \hat{g}$; **Parseval:** $\int |f|^2 dx = \frac{1}{2\pi} \int |\hat{f}|^2 dk$

Laplace Transform

$$\mathcal{L}\{f\}(s) = \int_0^{\infty} e^{-st} f(t) dt$$

- $\mathcal{L}\{1\} = 1/s$; $\mathcal{L}\{t^n\} = n!/s^{n+1}$; $\mathcal{L}\{e^{at}\} = 1/(s-a)$
- $\mathcal{L}\{\sin \omega t\} = \omega/(s^2 + \omega^2)$; $\mathcal{L}\{\cos \omega t\} = s/(s^2 + \omega^2)$
- $\mathcal{L}\{f''\} = s^2 F - sf(0) - f'(0)$; $\mathcal{L}\{tf\} = -F'(s)$; $\mathcal{L}\{f * g\} = FG$
- $\mathcal{L}\{\delta(t-a)\} = e^{-as}$; $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi/s}$; $\mathcal{L}\{t^{1/2}\} = \sqrt{\pi/(2s^{3/2})}$
- **Abel:** $\int_0^t g(\tau)(t-\tau)^{-1/2} d\tau \xrightarrow{\mathcal{L}} \sqrt{\pi/s} \cdot G(s)$

ODE Quick Reference

First-order types

- **Linear** $y' + Py = Q$: $\mu = e^{\int P dx}$, $y = \mu^{-1}(\int \mu Q dx + C)$
- **Separable:** $\int dy/g(y) = \int f dx$
- **Bernoulli** $y' + Py = Qy^n$: $v = y^{1-n}$ gives linear ODE
- **Exact** $M dx + N dy = 0$, $\partial_y M = \partial_x N$: find Φ s.t. $d\Phi = 0$
- **Riccati** $y' + y^2 + r = 0$: $y = -u'/u \Rightarrow u'' - ru = 0$

Second-order: constant coefficients

Char. eq. $r^2 + pr + q = 0$: real roots $\Rightarrow Ae^{r_1 x} + Be^{r_2 x}$; repeated $r \Rightarrow (A + Bx)e^{rx}$; complex $\alpha \pm i\beta \Rightarrow e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

Euler–Cauchy: $x^2 y'' + ax y' + by = 0$

Try $y = x^m$: $m(m-1) + am + b = 0$. Same three cases as above (replace x by x^m and βx by $\beta \ln x$).

Variation of parameters

$$y_p = -y_1 \int \frac{y_2 f}{W} dx + y_2 \int \frac{y_1 f}{W} dx, \quad W = y_1 y_2' - y_1' y_2$$

Wronskian – Abel: $W(x) = W(x_0) \exp(-\int_{x_0}^x P d\xi)$; constant if $P = 0$.

Leibniz rule: $\frac{d}{dx} \int_{a(x)}^{b(x)} g(x, y) dy = g(x, b) b' - g(x, a) a' + \int_{a(x)}^{b(x)} \partial_x g dy$

Convert to SL form: For $p_0 y'' + p_1 y' + p_2 y + \lambda w_1 y = 0$, multiply by $F = \exp(\int \frac{p_1 - p_0'}{p_0} dx)$; gives $p = F p_0$, $q = F p_2$, $w = F w_1$.

Bessel Functions

Bessel eq.: $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$. Solutions J_ν, Y_ν ; $Y_\nu \rightarrow \infty$ at 0.

$J_0(0) = 1$; $J_n(0) = 0$, $n \geq 1$. Zeros of J_0 : $\approx 2.405, 5.520, 8.654$; J_1 : $\approx 3.832, 7.016, 10.173$.

Recurrence: $J_{\nu-1} + J_{\nu+1} = \frac{2\nu}{x} J_\nu$; $[x^\nu J_\nu]' = x^\nu J_{\nu-1}$; $[x^{-\nu} J_\nu]' = -x^{-\nu} J_{\nu+1}$

Orthogonality (weight r): $\int_0^a J_\nu(\frac{j_{\nu,m}}{a} r) J_\nu(\frac{j_{\nu,n}}{a} r) r dr = \frac{a^2}{2} [J_{\nu+1}(j_{\nu,n})]^2 \delta_{mn}$

SL form: $-(rX')' + \frac{\nu^2}{r} X = \lambda r X$; $p = w = r$, $q = \nu^2/r$

Legendre Polynomials

Eq.: $(1-x^2)y'' - 2xy' + n(n+1)y = 0$, $x \in [-1, 1]$

$P_0 = 1$; $P_1 = x$; $P_2 = \frac{1}{2}(3x^2 - 1)$; $P_3 = \frac{1}{2}(5x^3 - 3x)$; $P_4 = \frac{1}{8}(35x^4 - 30x^2 + 3)$

Recurrence: $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$

Orthogonality: $\int_{-1}^1 P_m P_n dx = \frac{2}{2n+1} \delta_{mn}$; **Rodrigues:**

$$P_n = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n; \text{ spherical: } u = \sum A_n r^n P_n(\cos \theta)$$

Definite Integrals

Gaussian / Exponential

$$\begin{aligned} \int_0^\infty e^{-ax^2} dx &= \frac{\sqrt{\pi}}{2\sqrt{a}}; \quad \int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\pi/a}; \\ \int_0^\infty x^n e^{-ax} dx &= n!/a^{n+1} \\ \int_0^\infty e^{-ax} \sin bx dx &= \frac{b}{a^2+b^2}; \quad \int_0^\infty e^{-ax} \cos bx dx = \frac{a}{a^2+b^2}; \\ \int_0^\infty e^{-a^2 x^2} \cos bx dx &= \frac{\sqrt{\pi}}{2a} e^{-b^2/(4a^2)} \end{aligned}$$

Sinc & rational

$$\begin{aligned} \int_0^\infty \frac{\sin ax}{x} dx &= \frac{\pi}{2} \text{sgn}(a); \quad \int_0^\infty \frac{x \sin ax}{x^2+b^2} dx = \frac{\pi}{2} e^{-ab}; \\ \int_0^\infty \frac{\cos ax}{x^2+b^2} dx &= \frac{\pi}{2b} e^{-ab}, \quad a, b > 0 \end{aligned}$$

Wallis / Beta / Gamma

$$\begin{aligned} \Gamma(n+1) &= n!; \quad \Gamma(\tfrac{1}{2}) = \sqrt{\pi}; \quad \Gamma(x+1) = x\Gamma(x); \quad \text{B}(m, n) = \\ &\Gamma(m)\Gamma(n)/\Gamma(m+n); \quad \int_0^{\pi/2} \sin^{2n} x dx = \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} \end{aligned}$$

Integration by parts

$$\begin{aligned} \int x \sin \alpha x dx &= \frac{\sin \alpha x}{\alpha^2} - \frac{x \cos \alpha x}{\alpha}; \quad \int x \cos \alpha x dx = \frac{\cos \alpha x}{\alpha^2} + \frac{x \sin \alpha x}{\alpha} \\ \int x^2 \sin \alpha x dx &= \left(\frac{2}{\alpha^2} - \frac{x^2}{\alpha^2}\right) \cos \alpha x + \frac{2x}{\alpha^2} \sin \alpha x \\ \int_0^L x \sin \frac{n\pi x}{L} dx &= \frac{(-1)^{n+1} L^2}{n\pi}; \quad \int_0^L x \cos \frac{n\pi x}{L} dx = \\ &\frac{[(-1)^n - 1] L^2}{(n\pi)^2}, \quad n \geq 1 \end{aligned}$$

Delta Function & Heaviside

- **Sifting:** $\int f(x) \delta(x-a) dx = f(a); \quad \int f \delta'(x-a) dx = -f'(a); \quad \int f \delta''(x-a) dx = f''(a)$
- **Composition:** $\delta(g(x)) = \sum_i \frac{\delta(x-x_i)}{|g'(x_i)|}; \quad \delta(ax) = \frac{1}{|a|} \delta(x); \quad x \delta'(x) = -\delta(x)$
- **Heaviside:** $H'(x) = \delta(x); \quad \frac{d}{dx} H(g) = \delta(g) g'$
- **Jump:** $u'' = f + A \delta(x-x_0) \Rightarrow u'(x_0^+) - u'(x_0^-) = A$
- **Representations:** $\delta(x) = \lim_{\epsilon \rightarrow 0} \frac{e^{-x^2/\epsilon^2}}{\sqrt{\pi} \epsilon} = \frac{1}{2\pi} \int e^{ikx} dk$

Calculus of Variations

Euler-Lagrange: $I[y] = \int_a^b F(x, y, y') dx; \quad F_y - \frac{d}{dx} F_{y'} = 0$

Beltrami (F indep. of x): $F - y' F_{y'} = C$

Euler-Poisson (if F has y''): $F_y - \frac{d}{dx} F_{y'} + \frac{d^2}{dx^2} F_{y''} = 0$

Natural BC: $F_{y'}|_a = 0$ and/or $F_{y'}|_b = 0$ (for free endpoints)

Transversality (endpoint on ψ): $(F + (\psi' - y') F_{y'})| = 0$

Isoperimetric: minimize $I[y]$ s.t. $J[y] = \ell$: use $F^* = F + \mu G$

2D: $I[u] = \iint F(x, y, u, u_x, u_y) dA; \quad F_u - \partial_x F_{u_x} - \partial_y F_{u_y} = 0$

Steps:

1. Write E-L; use Beltrami if $\partial F / \partial x = 0$; Euler-Poisson if y'' in F
2. Isoperimetric: replace $F \rightarrow F^* = F + \mu G$; apply E-L to F^*
3. Solve ODE; apply endpoint / natural / transversality BCs
4. Determine μ from constraint $J[y] = \ell$

PDE Classification & Canonical Forms

For $Au_{xx} + Bu_{xy} + Cu_{yy} + \dots = 0$: $\Delta = B^2 - 4AC$

$\Delta > 0$: **Hyperbolic** ($u_{\xi\eta} = \Phi$); $\Delta = 0$: **Parabolic** ($u_{\eta\eta} = \Phi$); $\Delta < 0$: **Elliptic** ($u_{\xi\xi} + u_{\eta\eta} = \Phi$)

Characteristics: $A(dy)^2 - B dx dy + C(dx)^2 = 0$

1st-order method of characteristics: $au_x + bu_y = c$:

$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c}$$

Steps (canonical form):

1. Compute Δ ; classify type
2. Char. slopes: $A\lambda^2 - 2B\lambda + C = 0$; integrate $\rightarrow \xi(x, y), \eta(x, y)$
3. Transform: chain-rule for $u_x, u_{xx}, u_{xy}, u_{yy}$
4. Substitute $\rightarrow$ canonical form; integrate; apply Cauchy data

Wave Equation

d'Alembert (full line): $u = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$

Eigenfunction expansion (rod $[0, L]$, Dir-Dir): $\lambda_n = n\pi/L, \quad \omega_n = c\lambda_n, \quad u = \sum [A_n \cos \omega_n t + B_n \sin \omega_n t] \sin \frac{n\pi x}{L};$
 $A_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx, \quad B_n = \frac{2}{n\pi c} \int_0^L g \sin \frac{n\pi x}{L} dx$

Duhamel forcing: $u''_n + \omega_n^2 u_n = f_n(t), \quad f_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$

Resonance: forcing at $\Omega = \omega_n = cn\pi/L \Rightarrow$ secular growth $\sim t \sin \omega_n t$

2D rectangular membrane: $\phi_{nm} = \sin \frac{n\pi x}{L_x} \sin \frac{m\pi y}{L_y},$
 $\omega_{nm} = c\pi \sqrt{(n/L_x)^2 + (m/L_y)^2};$ square: $\omega_{nm} = \omega_{mn}$ (degenerate)

Steps (bounded):

1. Ansatz $u = X(x)T(t)$; spatial BVP $\rightarrow$ eigenpairs (λ_n, X_n)
2. Temporal ODE: $T''_n + \omega_n^2 T_n = F_n(t)$; Duhamel if forced
3. ICs $\rightarrow A_n, B_n$; check resonance $\Omega = \omega_n$

Heat Equation

Fundamental solution (full line): $u = \frac{1}{\sqrt{4\pi\kappa t}} \int_{-\infty}^\infty f(s) e^{-(x-s)^2/(4\kappa t)} ds$

Energy method (uniqueness): $E(t) = \frac{1}{2} \int_0^L u^2 dx; \quad \dot{E} = -\kappa \int_0^L u_x^2 dx \leq 0$

Special substitutions: advection $v_d u_x$: frame $\xi = x - v_d t$; reaction bu : set $u = e^{bt} w$

Steps (bounded domain):

1. Steady state u_s : solve $u_{s,xx} + \text{source} = 0$ with original BCs
2. Transient $w = u - u_s$: homogeneous BCs, IC $w_0 = u_0 - u_s$
3. Separate $w = X(x)T(t)$; eigenvalue problem $\rightarrow \lambda_n, X_n$
4. $T'_n + \lambda_n \kappa T_n = q_n(t)$; IF $= e^{\lambda_n \kappa t}$
5. Assemble $u = u_s + \sum c_n e^{-\lambda_n \kappa t} X_n$
6. Semi-infinite: Laplace transform in t (or Fourier in x); invert via erfc

Duhamel (heat): $u_t - \kappa u_{xx} = f$, zero IC, homog. BCs: $u = \int_0^t S(x, t-\tau; \tau) d\tau$

Laplace / Poisson

Poisson integral (disk $r < R$): $u(r, \theta) = \frac{R^2 - r^2}{2\pi} \int_0^{2\pi} \frac{f(\phi) d\phi}{R^2 - 2Rr \cos(\theta - \phi) + r^2};$ series: $u = \frac{a_0}{2} + \sum (\frac{r}{R})^n (a_n \cos n\theta + b_n \sin n\theta)$

Exterior ($r \geq R$): drop r^n , keep r^{-n} : $u = A_0 + \sum r^{-n} (A_n \cos n\theta + B_n \sin n\theta)$

Rectangle: $\sin(n\pi x/L)$ in x ; $\sinh/\cosh(n\pi y/L)$ in y (or transposed)

Neumann compatibility: $\oint_{\partial\Omega} g d\sigma = 0$; solution unique up to constant

Sector: $0 < r < R, \quad 0 < \theta < \alpha$, Dir in θ : $\nu = n\pi/\alpha$; $u = \sum A_n r^\nu \sin \nu\theta$ (non-integer ν if $\alpha \neq k\pi$)

Half-plane ($y > 0$): $u(x, y) = \frac{y}{\pi} \int_{-\infty}^\infty \frac{f(s) ds}{(x-s)^2 + y^2}$

Steps:

1. Coords: polar (disk/sector/annulus), Cartesian (rectangle)
2. Poisson: split $u = u_p + u_h$; $\nabla^2 u_p = S$ (often radial)
3. Separate; angular part $\rightarrow \nu$ ($= n$ for full disk; $= n\pi/\alpha$ for sector)
4. Regularity: drop $r^{-\nu}$ at origin, r^ν at ∞
5. Match BC: Fourier expand; equate coefficients

Green's Functions

Definition: $LG(x, \xi) = \delta(x - \xi)$ (same BCs); solution: $u(x) = \int G(x, \xi) f(\xi) d\xi$

Closed form (Wronskian method): $G(x, \xi) = \frac{y_1(x_<)y_2(x_>)}{p(\xi)W(\xi)}$, $x_< = \min / \max(x, \xi)$; y_1 : left BC, y_2 : right BC

Jump condition: $G_x(\xi^+) - G_x(\xi^-) = 1/p(\xi)$ (for $L = -(py')' + q$)

Templates:

- $-y'' + k^2 y = f$ on $[0, \pi]$ (Dir): $G = \frac{\sinh kx_< \sinh k(\pi - x_>)}{k \sinh k\pi}$
- $-y'' - k^2 y = f$ on $[0, L]$ (Dir, $kL \neq n\pi$): $G = \frac{\sin kx_< \sin k(L - x_>)}{k \sin kL}$
- Robin left BC $y'(0) = h_1 y(0)$: $y_1 = \cos kx + \frac{h_1}{k} \sin kx$

Eigenfunction expansion: $G = \sum_n \frac{\phi_n(x)\phi_n(\xi)}{\lambda_n}$; modified GF (zero eigenvalue): $LG_m = \delta - \phi_0^2$; need $\int f \phi_0 dx = 0$

Steps:

1. Solve $Ly = 0 \rightarrow y_1$ (left BC), y_2 (right BC)
2. $G = y_1(x_<)y_2(x_>)/(p(\xi)W)$
3. Verify: continuity, jump $[G_x]_\xi = 1/p(\xi)$, symmetry $G(x, \xi) = G(\xi, x)$
4. $u(x) = \int G(x, \xi) f(\xi) d\xi$

Integral Equations

Fredholm 2nd kind: $\varphi = f + \lambda \int_a^b K(x, t) \varphi(t) dt$

Degenerate kernel $K = \sum g_i(x) h_i(t)$: set $c_i = \int h_i \varphi dt$; $\varphi = f + \lambda \sum c_i g_i$; substitute back $\rightarrow$ linear system $(I -$

$\lambda M)c = b$, $M_{ij} = \int h_i g_j dt$; eigenvalues: $\det(I - \lambda M) = 0$

Fredholm alternative: if $\lambda = \lambda^*$, solution exists iff $\langle f, \psi^* \rangle = 0$

Neumann series (small λ): $\varphi = \sum \lambda^n K^{(n)} f$; converges for $|\lambda| \|K\|_{L^2} < 1$

Volterra 2nd kind: $u = f + \lambda \int_a^x K(x, t) u(t) dt$; differentiate (Leibniz) $\rightarrow$ ODE; IC: $u(a) = f(a)$; convolution $K(x - t)$: Laplace $\rightarrow \hat{u} = \hat{f}/(1 - \lambda \hat{K})$; Abel $K = (x - t)^{-\alpha}$: use $\mathcal{L}\{t^{-\alpha}\} = \Gamma(1 - \alpha)/s^{1-\alpha}$

Steps (degenerate kernel):

1. Write $K = \sum \phi_i(x) \psi_i(t)$; set $c_i = \int \psi_i \varphi dt$
2. $\varphi = f + \lambda \sum c_i \phi_i$; substitute back $\rightarrow (I - \lambda M)c = b$
3. Solve; eigenvalues: $\det(I - \lambda M) = 0$

Perturbation Theory

Regular: expand $x = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots$; substitute; collect equal powers of ϵ .

Singular / dominant balance: when roots blow up as $\epsilon \rightarrow 0$:

1. Assume $x \sim A\epsilon^{-\alpha}$; substitute; balance two dominant terms $\rightarrow \alpha$
2. Rescale $x = \epsilon^{-\alpha}(1 + \delta)$; solve for correction δ
3. Corrections may involve non-integer powers of ϵ

Sturm–Liouville Theory

Regular SL: $-(py')' + qy = \lambda wy$ on $[a, b]$

- Eigenvalues $\lambda_1 < \lambda_2 < \dots \rightarrow +\infty$, all real
- Eigenfunctions orthogonal w.r.t. w : $\int_a^b \phi_m \phi_n w dx = 0$, $m \neq n$
- Completeness: $f = \sum c_n \phi_n$, $c_n = \langle f, \phi_n \rangle_w / \|\phi_n\|_w^2$
- Robin BC transcendental eq.: $(h_1 + h_2) \cos kL = (k + \frac{h_1 h_2}{k}) \sin kL$
- SA boundary form: $\langle u, Lv \rangle - \langle Lu, v \rangle = [p(uv' - u'v)]_a^b = 0$
- Convert to SL: multiply $p_0 y'' + p_1 y' + p_2 y + \lambda w_1 y = 0$ by $F = e^{\int (p_1 - p_0')/p_0 dx}$